



Preparation of DNA Libraries and Loading on the NextSeq 1000/2000

Influenza Division, Virology, Surveillance and Diagnosis Branch, Genomics and Diagnostic Team

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1.0 Purpose

- 1.1 This protocol describes how the Influenza DNA library is prepared before loading on the NextSeq 1000/2000 and the procedures for use of the Illumina NextSeq 1000/2000 sequencing platform.

2.0 Definitions

- 2.1 RSB with Tween 20 – Illumina resuspension buffer
- 2.2 PhiX – a quality control for cluster generation, sequencing, and alignment; and a calibration control for crosstalk matrix generation, phasing, and pre-phasing.

3.0 Critical Equipment

- 3.1 Illumina NextSeq 1000/2000
- 3.2 Microcentrifuge
- 3.3 Vortex
- 3.4 Adjustable pipettes – P10, P100, and P1000

4.0 Materials

4.1

NextSeq 1000/2000 Series Reagent Kit 300 cycles	
Flow Cell	Product Number
NextSeq 1000/2000 P1 XLEAP-SBS Reagent Kit	20100982
NextSeq 1000/2000 P2 XLEAP-SBS Reagent Kit	20100985

- 4.2 PhiX Control v3 Illumina, FC-110-3001

5.0 Controls

- 5.1 PhiX – a quality control for cluster generation, sequencing, and alignment; and a calibration control for cross-talk matrix generation, phasing, and pre-phasing.

6.0 Safety Precautions

- 6.1 Adhere to the safety guidelines provided in the Biosafety in Microbiological and Biomedical Laboratories (BMBL) and follow all established site-specific safety procedures, including wearing proper personal protective equipment (PPE).



7.0 **The day before anticipated loading of the NextSeq 1000/2000**

7.1 Thaw XLEAP-SBS cartridge by refrigerator or room temperature method:

7.1.1 **Refrigerator**

- One day before the anticipated run, remove the cartridge from the box but do not open the silver foil bag.
- Position the cartridge with label facing up.
- Thaw at room temperature for 6 hours.
- Move the cartridge with the label facing up to 2-8°C.
- Thaw in the refrigerator for 16 hours. Do not exceed 72 hours.

7.1.2 **Thaw at Room Temperature**

- The evening before the anticipated run, remove the cartridge from the box but do not open the silver foil bag.
- Position the cartridge with the label facing up.
- Thaw at room temperature for 12 hours. Do not exceed 16 hours.

8.0 **NextSeq 1000/2000 Procedures**

8.1 **Prepare the Flow Cell**

8.1.1 Remove a new flow cell from 2-8°C storage.

8.1.2 Set unopened package aside at room temperature for 10-15 minutes to prevent condensation when opening.

8.2 **Preparation of 750 pM PhiX v3 Control Stock**

8.2.1 At room temperature, combine the following volumes of reagents to dilute the 10nM PhiX Control to 750 pM:

<u>Reagent</u>	<u>Volume (µl)</u>
10 nM PhiX Control	1.5 µl
RSB with Tween 20	18.5 µl
Total	20.0 µl (2 nM PhiX Stock)

8.2.2 Vortex briefly to mix the 750 pM PhiX solution.

8.2.3 Briefly spin the template solution.

8.3 **Normalization of the NDA Libraries**

8.3.1 Normalize the pooled DNA library to 2nM using the concentration from the Qubit assay and genome size determination from a fragment analyzer.

NOTE: If needed, pooled libraries can be diluted to ~1.0 ng/µl or loading concentrations can be adjusted for optimal cluster density.

8.3.2 Dilute DNA libraries to 275 pM:

Reagent	Volume (μl)
Pooled 2 nM DNA library	3.3 μl
RSB with Tween 20	20.7 μl
Total	24.0 μl

8.3.3 Vortex briefly to mix the solution.

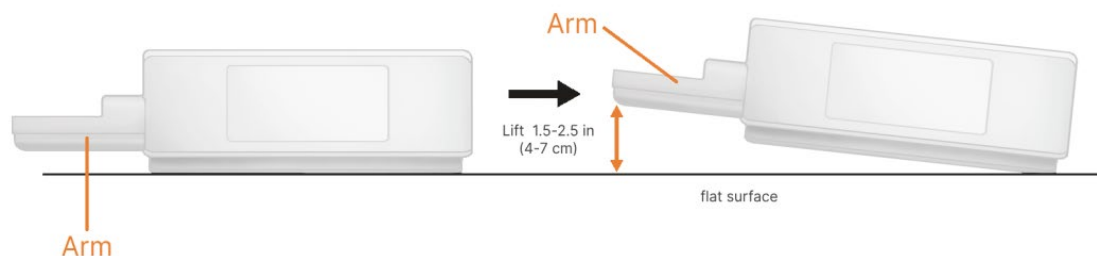
8.3.4 Store mixture on ice until ready to use.

8.4 **Prepare the Cartridge and Load Flow Cell**

8.4.1 Open the cartridge bag and discard.

8.4.2 Slowly invert the cartridge side to side 10 times to mix reagents.

8.4.3 Place the cartridge on a flat surface and lift the cartridge by the arm 1.5-2.5 inches and release onto the bench 5 times.



8.4.4 This reduces the bubbles and stabilizes cartridge components.

8.4.5 Open the foil package containing the flow cell.

8.4.6 Hold the flow cell by the gray tab with the label on the tab facing up.

8.4.7 Push to insert the flow cell into the slot on the front of the cartridge. The flow cell should click when in place.

8.4.8 Pull back and remove the gray tab to expose the flow cell.

Note: The gray tab can be recycled.

8.5 **Load Libraries into Cartridge**

8.5.1 Mix the following:

Reagent	Volume (μl)
Pooled 275 pM DNA library	19.0 μl
750 pM PhiX (5% of total)	1.0 μl
Total	20.0 μl

8.5.2 Using a new P1000 pipette tip, pierce the **Library** reservoir.

8.5.3 Discard the pipette tip.

8.5.4 Add 20 μl of the diluted library with PhiX to the bottom of the reservoir.



8.6 **Begin Sequencing Run and Load Consumables**

- 8.6.1 From the control software menu, select **Disk Management**.
- 8.6.2 Delete all runs except for the previous run.
 - This deletes from the local run folder.
- 8.6.3 Check that previous sequencing runs have been deleted from the instrument hard drive in the output folder; only keep the previous run on the instrument.
- 8.6.4 Select **Start**.
- 8.6.5 Sign in with local account.
- 8.6.6 Select **Choose** under Start with Sample Sheet.
- 8.6.7 Navigate to the sample sheet generated by Clarity or manually created.
- 8.6.8 Select **Review**.
- 8.6.9 Review the run parameters.
- 8.6.10 On the review run screen, select Load.
 - Reagent door opens.
- 8.6.11 Place the cartridge onto the tray with the label facing up and the flow cell inside the instrument.
- 8.6.12 Push cartridge until it locks into seat.
- 8.6.13 Select **Close**.
- 8.6.14 Select **Sequence**.
- 8.6.15 Pre-run checks take about 15 minutes.
- 8.6.16 After pre-runs check is completed, the instrument will start auto automatically.

8.7 **Monitoring the Run**

- 8.7.1 Monitor the following metrics on the Sequencing or Runs screen:
 - Run metrics are not available until cycle 26 of Read 1.
 - **% ≥ Q30** – The expected number of base calls with a Q-score ≥ 30
 - **Projected yield** – the expected number of bases called for the run
 - **Total reads PF** – the number of paired-end reads passing filter (in millions)
 - **Total % demux** – the percentage of PF reads demultiplexed for the run

8.8 **Completed Run**

- 8.8.1 When the run is complete, select **Eject Cartridge**.
- 8.8.2 Remove the flow cell from the cartridge.
- 8.8.3 Disassemble and discard the flow cell in the autoclave bin.
- 8.8.4 Remove the drain plug under the Illumina log on the side of the cartridge.
- 8.8.5 Drain contents into the hazardous waste container.



8.8.6 Disassemble and discard the cartridge in the autoclave bin.

8.8.7 Recycle all plastic components.

8.8.8 Select **Close**.

8.9 **Maintenance**

8.9.1 Replace the air filter every 6 months.

8.9.2 If cartridge leaks, clean the cartridge tray with 10% bleach, followed by 70% ethanol.

8.9.3 Preventive maintenance is performed by Illumina yearly.

9.0 **Related Procedures**

9.1 LP-516 DNA Quantification with Qubit ds DNA High Sensitivity Kit

9.2 LP-519 Manual Procedure Illumina DNA Prep Library SOP

9.3 LP-520 QIAxcel Sample Preparation and Quality Control Procedure

10.0 **References**

10.1 Illumina NextSeq 1000/2000® System User Guide (ILLUMINA Part # 200027171 v04)
[https://support-
docs.illumina.com/IN/NextSeq10002000/Content/IN/FrontPages/NextSeq10002000.htm](https://support-docs.illumina.com/IN/NextSeq10002000/Content/IN/FrontPages/NextSeq10002000.htm)

10.2 [Biosafety in Microbiological and Biomedical Laboratories \(BMBL\), current edition](#)

11.0 **Attachments - None**